ABC DIGITAL SERVICES

Corporate VPN Connectivity and Split-Tunnel Troubleshooting

Approved IT Knowledge Article Pack

Document ID	Version	Status	Effective date
ADS-KB-NET-002	1.0	Approved for capstone use	06 Nov 2026

Knowledge owner	Review cycle	Classification	Issue category
Network Operations	Every 6 months	Internal	Network & VPN

Fictional organization and sample operating procedures

How to use this article

This document is authoritative only for the fictional ABC Digital Services (ADS) capstone environment. The assistant may explain and recommend these steps, but must not execute privileged actions, approve expenditure, change access, or close a case on the user's behalf.

Retrieval metadata	Value
Document ID	ADS-KB-NET-002
Category	Network & VPN
Products / assets	Always-On VPN, Error 809, IKEv2, certificate auth, split-tunnel, internal DNS
Platform	Windows 11, macOS Sequoia, Cisco AnyConnect / Windows Always-On VPN
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Superseded by	None

Safety boundary: Never request passwords, MFA codes, recovery keys, private keys, or full payment-card data. Stop and escalate if there is smoke, swelling, electric shock, suspected compromise, or possible data loss.

1. Resolve VPN Error 809 (UDP 500/4500 blocked)

Field	Guidance
Common user phrases	"VPN error 809", "network connection between computer and VPN server could not be established"
Default priority	P2 High when blocking remote work; P3 Normal otherwise
Objective	Restore IKEv2 tunnel negotiation or fail over to TLS 443 cleanly.
Minimum intake	Asset tag; user ISP/hotel network; VPN gateway hostname; error code; local firewall status.

Procedure

1. Verify the user has working external internet connectivity by testing public HTTPS sites.

2. Check if the user is on a restricted guest Wi-Fi or mobile hotspot blocking UDP ports 500 and 4500 required for IKEv2.

3. Run network diagnostics against vpn-gw-01.ads.local on TCP port 443 and verify gateway latency.

4. Instruct user to switch the ADS VPN client profile from 'Automatic (IKEv2)' to 'SSTP / TLS 443 Fallback'.

5. Flush local DNS resolver cache (`ipconfig /flushdns`) and re-test internal intranet reachability.

Resolution and escalation

Outcome	Criteria
Resolve when	VPN tunnel establishes, internal corp DNS resolves, and latency is below 150ms.

Outcome	Criteria
Escalate when	Both IKEv2 and TLS 443 fail; gateway health check reports DEGRADED; user device certificate is revoked.
Evidence to record	Gateway probe result, protocol used, ISP type, DNS test confirmation.
Forbidden actions	Instructing user to disable Windows Defender Firewall globally; sharing static VPN pre-shared keys.

2. Renew expired machine certificate for VPN authentication

Field	Guidance
Common user phrases	"VPN certificate expired", "no valid client certificate found", "Error 798"
Default priority	P3 Normal
Objective	Renew enterprise PKI machine certificate via Intune/Auto-Enrollment.
Minimum intake	Asset tag; certificate thumbprint expiry date; current network connection.

Procedure

1. Connect via the temporary TLS bootstrap profile or corporate office LAN.

2. Trigger Intune Company Portal sync or run `certutil -pulse` in a standard terminal.

3. Verify the new ADS-Endpoint-Auth certificate appears under Local Machine Personal Certificates.

4. Disconnect bootstrap profile and reconnect via primary Always-On VPN tunnel.

Resolution and escalation

Outcome	Criteria
Resolve when	New PKI certificate is valid and primary VPN connects automatically.
Escalate when	Device is dropped from Active Directory trust or Intune MDM enrollment is broken.
Evidence to record	Certificate expiration date, Intune sync timestamp, tunnel status.
Forbidden actions	Exporting private keys (.pfx) across machines; installing untrusted root certificates.

General ticket handling rules

Priority	Use when
P1 Critical	Widespread outage, active security threat, severe safety risk, or executive-critical service unavailable. Page the major-incident or security team immediately.
P2 High	User cannot perform time-critical work and no practical workaround exists. Target assignment within 30 minutes.

Priority	Use when
P3 Normal	Single-user degradation or standard fulfillment request. Target assignment within 4 business hours.
P4 Low	Information request, planned change, or non-urgent accessory request. Target assignment within 1 business day.

Standard ticket record

Record requestor, business justification, device or application, asset tag where applicable, location, symptoms, error text, start time, impact, troubleshooting attempted, approvals, evidence, result, and next owner. Do not paste secrets or unnecessary personal data.

Knowledge quality and review

The knowledge owner reviews this article every six months and after any major process, product, security, or supplier change. Superseded versions must be excluded from retrieval. Conflicts are resolved in favor of the newest approved article; otherwise the assistant must disclose the conflict and recommend escalation.

Reference basis

• Secure AI Troubleshooting Assistant BRD v1.0: approved content, metadata, ordered guidance, citations, verification, safe fallback, and escalation.

• NIST SP 800-61 Rev. 3 (2025): Detect, Respond, Recover, and continuous improvement concepts for security-related incidents.

• Microsoft Intune and Windows Autopilot documentation: managed application deployment, device enrollment, assignment, and lifecycle practices.

• ITIL-aligned service management practices: incident, request, change, asset, and knowledge management concepts.